

Lava — Jackett / Comprehensive-QA Program Worklog

Source request: docs/requests/improvements/lava_request_jackett.md (credentials redacted out of it 2026-06-08 per §6.H). This worklog is the durable record of the program (§6.S spirit). It carries **no credentials**. Started: 2026-06-08.

0. Program scope (from the operator request)

1. Fetch/pull latest constitution + all submodules; verify full compliance.
2. Add **all supported test types** that, per supported provider: onboard → search (that provider only) → browse results → validate/verify content → **confirm a real, working download option**: a valid (non-broken) download link, a really-downloaded **non-empty valid .torrent**, OR a **100%-valid magnet**. Replace manual human testing.
3. Enumerate + test **every real-human use case**, variations, edge cases.
4. **HelixQA**: already a submodule — build comprehensive test banks + **autonomous per-provider QA sessions** (browse/search/obtain-download). Provide a **Claude↔HelixQA model bridge** (LLM/Vision) + real-time monitoring.
5. **Jackett**: deep analysis; ideally submodule + reference (decision pending — language boundary); else port providers, all fully test-covered.
6. Use the **4 provider credentials** (RuTracker, RuTor, IPTorrents, NNMCclub); incorporate unsupported ones fully. No leaks via git/logs.
7. **Full decoupling** — providers easily pluggable; future **Android-extension-APK** providers.
8. Maximally extend all docs/diagrams/SQL/architecture.

- 9. Regular commit+push of all submodules + main to all upstreams (GitHub+GitLab per §6.W).
- 10. **Maximal test coverage**, all test types, every line. **Real evidence only** — **zero bluffs**.

1. Done (verifiable)

- **§6.H credential safety**: redacted 4 cred lines from the request md (untracked, but docs/ not gitignored). Leak scan: nobody85perf* exists in git only in already-recorded forensic files (2026-05-17 incident JSON + CLAUDE/CHANGELOG anchors). No new leak.
- **.env** (gitignored): added RUTOR_*, IPTORRENTS_*; RuTracker/NNMClub already matched provided values. Honest header note added; **RuTracker password remains UNROTATED** (standing §6.H operator action).
- **.env.example** (tracked): added RUTOR_* + IPTORRENTS_* placeholders (§6.R).
- **Submodules fetched**. Divergence: constitution -12, helixqa -16, challenges -3, containers -1, security -2; rest current. **Pin bumps deferred** until recon agents (reading current pins) finish.

2. Wave-1 recon findings

Agent 1 — Provider inventory (COMPLETE)

6 providers today:

Provider	Auth	.torrent	Magnet	§6.E status
rutracker	CAPTCHA_LOGIN	✓	async-only (sync returns null)	PARTIAL
rutor	FORM_LOGIN	✓	sync from search HTML	PARTIAL (sync getMagnetLink null)
kinozal	FORM_LOGIN	✓	null (but present in	CONFIRMED GAP / bluff-risk

Provider	Auth	.torrent	Magnet	§6.E status
			search HTML)	
nnmclub	FORM_LOGIN	✓	null (but present in search HTML)	CONFIRMED GAP / bluff-risk
archiveorg	NONE	HTTP file (not torrent)	n/a	correct by design
gutenberg	NONE	HTTP e-book (not .torrent)	n/a	TORRENT_DOWNLOAD label mismatch

Registration: `DefaultTrackerRegistry.register(factory)` (`core/tracker/client/di/TrackerClientModule.kt:270-275`) → `LavaTrackerSdk` → `factory.create(config)`. Capability gate: `TrackerClient.getFeature(KClass)` reads `descriptor.capabilities` (§6.E).

Download URLs (cited):

- rutor `https://d.rutor.info/download/{id}` (`RuTorDownload.kt:38`)
- kinozal `https://kinozal.tv/download.php?id={id}` (`KinozalDownload.kt:29`)
- nnmclub `https://nnmclub.to/forum/download.php?id={id}` (`NnmclubDownload.kt:29`)
- archiveorg `https://archive.org/download/{id}/{file}` (`ArchiveOrgDownload.kt:48`)
- gutenberg Gutendex `https://gutendex.com/books/{id}/` (`GutenbergDownload.kt:44`)

§6.R: all 6 base URLs hardcoded literals in descriptors (`rutracker:13`, `rutor:25`, `kinozal:19`, `nnmclub:19`, `archiveorg:23`, `gutenberg:24`). Clone-override path partially mitigates (download impls accept ctor `baseUrl` override) but descriptor mirror lists are static.

Derived workable items (evidence-backed):

- W1. Wire kinozal magnet through (data in `KinozalSearchParser.kt:52`) + regression test (falsifiable).
- W2. Wire nnmclub magnet through (`NnmclubSearchParser.kt:52`) + regression test.

- W3. Resolve gutenbergl capability label (TORRENT_DOWNLOAD → HTTP_DOWNLOAD or correct semantics).
- W4. Resolve rutracker/rutor sync-magnet-null vs declared MAGNET_LINK (per §6.E).
- W5. §6.R base-URL config-driving (broad; coordinate with mirror manager).

Agent 2 — Test infrastructure (COMPLETE)

127 test files (3 Android unit + 45 Challenges + Go contract/e2e/integration/parity/load/qa + panoptic). **The operator's core ask has ~zero real coverage:**

- Deep Android flows (search → topic → download → bencode-validate → magnet-validate) are MISSING or were gutted to "shallow" by a **nav-compose 2.9.0 lifecycle bug** (C04-C08 + deep C11; see .lava-ci-evidence/sp3a-challenges/C4-thru-C8-2026-05-04-shallow-redesign.json).
- Go e2e /download/{id} returns a **synthetic** payload d8:announce10:fake-trackere (e2e_test.go:280) — validates wire contract, NOT real bencode/magnet.
- **GAP-5/6:** no bencode .torrent structural validation, no magnet-format validation anywhere.
- Creds injected via BuildConfig from .env (app/build.gradle.kts), gated by assumeTrue() (honest SKIP, not false PASS).

Agent 3 — HelixQA (COMPLETE)

Go autonomous-QA framework. **Claude↔HelixQA bridge ALREADY EXISTS:** pkg/llm/bridge_provider.go BridgedCLIProvider shells to the claude CLI as an LLM/Vision Provider (Chat/Vision/Name/SupportsVision). Vision: pkg/vision/cheaper, nav: pkg/visionnav. 4-phase autonomous session (setup→doc-verify→curiosity→report) w/ video + ADB nav + issue-detect + ticket-gen. Banks = YAML (≥30 floor). **Lava wires today:** qa/detector, qa/evidence, qa/validator, qa/ticket. **Missing:** LLM/vision/navigator(Q6 skipped) wiring, NO Lava-Android banks, no per-provider session multiplexing, no autonomous run against the Lava app.

Agent 4 — Jackett decision matrix (COMPLETE)

Jackett = C#/.NET proxy; tracker coverage = **Cardigann YAML** defs (src/Jackett.Common/Definitions/, schema v1-v11, ~500+ indexers, shared w/ Prowlarr/Indexers) interpreted by a C# engine. **No in-**

process cross-language reuse (CLR vs JVM/Go). Consumed via **Torznab HTTP** (/api/v2.0/indexers/<id>/results/torznab/api):
RSS+torznab:attr → <enclosure> .torrent URL + magneturl + infohash
per <item>. **License** = **GPL-2.0** (decisive):

- Option A run-as-service (Torznab sidecar) — GPL arm's-length SAFE; server-side only (no on-device APK); effort LOW; anti-bluff evidence EXCELLENT.
- Option B port interpreter+YAML to Kotlin/Go — on-device capable + APK-ready, but **GPL-2.0 contaminates the distributed APK** (Play-Store friction); effort HIGH; needs legal sign-off.
- Option C hybrid — A now for breadth + clean-room non-GPL interpreter later for on-device.

UNCONFIRMED/legal: exact YAML-only count; Prowlarr/Indexers exact license (GPL-family); vasic-digital policy on GPL submodule; browse/forum flows vs Torznab search-centric mapping.

Synthesized program direction

- **Required 4 trackers:** rutracker ✓, rutor ✓, nnmclub ✓ already native; **IPorrents** = **NEW native provider** (safe, on-device-ready, no GPL) — proceed regardless of Jackett choice.
- **Existing-provider hardening** (the heart of the request): validators + magnet §6.E fixes + real-network download/magnet verification + HelixQA banks — in flight, no operator gate.
- **Jackett mass-tracker breadth:** pending operator decision (A/B/C) — GPL + on-device-APK fork.
- **nav-compose 2.9.0** → bump to 2.9.1 (root cause confirmed; apply in integration pass; device-verify when Genymotion VM booted).
- **kinozal/nnmclub magnet §6.E bluffs:** fixes in flight (TDD + Bluff-Audit).

Agent 5 — Decoupling + APK feasibility (COMPLETE)

Clean compile-time plugin abstraction (PluginFactory / getFeature / 7 feature interfaces). **But hard-coupled registration:** TrackerClientModule.provideTrackerRegistry() (core/tracker/client/di/TrackerClientModule.kt:260-276) manually injects+registers each factory — no service-loader/reflection. To add a provider: new core/tracker/<id>/ module (descriptor+client+factory+feature files+fixtures+tests+Challenge) + edit settings.gradle.kts + edit that one registry method. **Android-extension-APK:** ~4-6wk refactor —

needs core/tracker/extension-api/ (Parcelable manifest + ContentProvider/AIDL IPC), secondary runtime registry, per-extension test duplication. Tachiyomi/Mihon model cited UNCONFIRMED.

Agent 6 — Constitution compliance (COMPLETE)

Governing: §6.D/6.E/6.G (coverage+capability honesty+provider E2E), §6.H/§11.4.10 (creds), §6.R (no-hardcode), §6.I/6.V/6.X/6.AE/6.AH (emulator-in-container/VM matrix), §6.W (mirrors), §6.S (CONTINUATION), §6.O (Crashlytics), §6.P (versioning), §11.4.6 (no-guessing), §11.4.76 (forbidden emulator cmds). **§6.AH-debt reportedly resolved 2026-06-06 via Genymotion VM + S23 Ultra** — but VM not live now (verified: adb 0 devices, :6555 closed). Per-new-provider checklist: capability-honesty unit test, §6.D behavioral test, §6.G auth-honesty + real-stack + falsifiability + Challenge, §6.R no-hardcode, per-AVD attestation on VM/device.

3. Known blockers (honest)

- **§6.AH-debt / §6.X-debt**: container/VM Android-emulator gate does NOT boot on this macOS host (no /dev/kvm, no HVF-in-container). Device-level Challenge evidence blocked here. Real download/magnet proof to be driven at Go-API + JVM + real-network layers (fetch real .torrent, assert non-empty valid bencode; validate magnet structure) — no bluffed device PASS.

4. Wave-2 — VERIFIED + COMMITTED (2026-06-08)

Independently re-verified by forced --rerun-tasks (not agent-trusted):
core:common:test + core:tracker:kinozal:test +
core:tracker:nmclub:test → BUILD SUCCESSFUL 35s;
TorrentFileValidatorTest 11/0/0, MagnetLinkValidatorTest 9/0/0,
KinozalDownloadTest 2/0/0, NmclubMagnetExposureTest 3/0/0. Spotless clean.

Committed to master (NOT yet pushed — batching with rutor fix):

- 776dc934 feat(core/common): bencode + magnet validation library (lava.common.torrent.
{Bencode,TorrentFileValidator,MagnetLinkValidator}).

- 6fa31ad2 fix(core/tracker/kinozal): §6.E magnet exposure via KinozalMagnetCache.
- 13703bc8 fix(core/tracker/nmclub): §6.E magnet exposure via NnmclubMagnetCache.

In flight: rutor magnet §6.E fix (same pattern), Jackett local-stack deep research (→ docs/qa/jackett-local-stack-research.md).

5. Operator decisions captured

- **Credentials** (all 5): in gitignored .env (RuTracker, RuTor, IPTorrents, NNMClub, **Kinozal updated to nobody85perf/01G...**). .env.example carries placeholders. NEVER committed.
- **Jackett = Option 1 (Torznab sidecar)** BUT must run as part of the **Lava local stack** (mDNS-discovered local API, same model as lava-api-go; NOT in the APK). Deep research first (in flight).
- **IPTorrents** = NEW native provider (the one of the 4 required trackers not yet supported).

6. Pending workable items (backlog)

- W4: rutracker/rutor sync-magnet §6.E (rutor fix in flight; rutracker has async GetMagnetLinkUseCase — verify honest).
- W3: gutenberg TORRENT_DOWNLOAD label vs HTTP e-book — resolve capability semantics.
- nav-compose 2.9.0→2.9.1 bump (apply + compile-verify; device-verify when Genymotion VM booted).
- Real-network per-provider download/magnet verification tests (the manual-testing replacement) using the new validators, gated -PrealTrackers, real .env creds.
- IPTorrents native provider (after Jackett/FlareSolverr research informs Cloudflare handling).
- HelixQA binary won't build — submodules/helixqa/go.mod has git conflict markers (L124/131/138); bump pin (always-track Q9) to fix.
- Jackett sidecar local-stack implementation (after research).
- Submodule pin bumps (constitution -12, challenges -3, containers -1, security -2) — deliberate, after current wave settles.
- Push wave-2+rutor batch to GitHub+GitLab (§6.W).